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Querying Causal Databases

- Introduction
- Building a Richer Atlas
- Multi-hop Causal Paths
- 3-hop Causal Chains
- Feedback Loops
- Controversial Claims
- Relation Distribution
- Combining Queries with Custom SQL
- Summary

Introduction

CSQL.jl provides a rich query API over causal databases. Beyond simple backbone and hub queries (covered in Getting Started), this vignette demonstrates:

- **Multi-hop causal paths** — chains like $A \rightarrow B \rightarrow C$
- **Feedback loops** — mutual influence (2-cycles)
- **Controversial claims** — edges with mixed directional evidence
- **Relation distribution** — what kinds of causal relationships dominate?

Building a Richer Atlas

We'll build a larger epidemiological causal atlas with feedback loops and conflicting evidence:

```
using CSQL
```

```
builder = AtlasBuilder()
```

```
# Core transmission chain
```

```
add_triple!(builder, "Pathogen transmissibility", "increases", "Transmission rate");
```

```

        doc_id="smith2023", score=0.92)
add_triple!(builder, "Contact rate", "increases", "Transmission rate";
        doc_id="jones2024", score=0.87)
add_triple!(builder, "Transmission rate", "increases", "Outbreak severity";
        doc_id="smith2023", score=0.95)
add_triple!(builder, "Outbreak severity", "increases", "Case fatality rate";
        doc_id="smith2023", score=0.73)

# Immunity feedback
add_triple!(builder, "Vaccination coverage", "reduces", "Host susceptibility";
        doc_id="chen2023", score=0.85)
add_triple!(builder, "Vaccination coverage", "increases", "Population immunity";
        doc_id="chen2023", score=0.91)
add_triple!(builder, "Population immunity", "reduces", "Host susceptibility";
        doc_id="patel2024", score=0.77)
add_triple!(builder, "Host susceptibility", "increases", "Transmission rate";
        doc_id="smith2023", score=0.80)

# Feedback loop: severity public health response
add_triple!(builder, "Outbreak severity", "increases", "Public health response";
        doc_id="who2024", score=0.88)
add_triple!(builder, "Public health response", "reduces", "Outbreak severity";
        doc_id="who2024", score=0.82)

# Healthcare
add_triple!(builder, "Healthcare capacity", "reduces", "Case fatality rate";
        doc_id="patel2024", score=0.88)
add_triple!(builder, "Outbreak severity", "reduces", "Healthcare capacity";
        doc_id="who2024", score=0.65)

# Controversial claim: does quarantine reduce transmission?
add_triple!(builder, "Quarantine", "reduces", "Transmission rate";
        doc_id="brown2023", score=0.78)

# Opposing evidence
add_triple!(builder, "Quarantine", "increases", "Transmission rate";
        doc_id="lee2024", score=0.45)

csql = connect_csql()
build!(builder, csql.db)
stats = statistics(csql)
println("Atlas: $(stats[:n_nodes]) nodes, $(stats[:n_edges]) edges, $(stats[:n_support]) sup
Atlas: 11 nodes, 14 edges, 14 support records

```

Multi-hop Causal Paths

Find 2-hop causal chains ($A \rightarrow B \rightarrow C$):

```
paths = causal_paths(csql; depth=2, limit=10)
println("Found $(length(paths)) 2-hop paths:")
for p in paths
    println(" $(p.a) -[$(p.r1)]→ $(p.b) -[$(p.r2)]→ $(p.c) (path_score=$(round(p.path_score)))")
end
```

Found 10 2-hop paths:

```
pathogen transmissibility -[INCREASES]→ transmission rate -[INCREASES]→ outbreak severity
transmission rate -[INCREASES]→ outbreak severity -[INCREASES]→ public health response (path_score=1.73)
contact rate -[INCREASES]→ transmission rate -[INCREASES]→ outbreak severity (path_score=1.73)
host susceptibility -[INCREASES]→ transmission rate -[INCREASES]→ outbreak severity (path_score=1.73)
quarantine -[REDUCES]→ transmission rate -[INCREASES]→ outbreak severity (path_score=1.73)
public health response -[REDUCES]→ outbreak severity -[INCREASES]→ public health response
outbreak severity -[INCREASES]→ public health response -[REDUCES]→ outbreak severity (path_score=1.73)
vaccination coverage -[INCREASES]→ population immunity -[REDUCES]→ host susceptibility (path_score=1.73)
transmission rate -[INCREASES]→ outbreak severity -[INCREASES]→ case fatality rate (path_score=1.73)
vaccination coverage -[REDUCES]→ host susceptibility -[INCREASES]→ transmission rate (path_score=1.73)
```

These multi-hop paths reveal indirect causal mechanisms — e.g., how pathogen transmissibility ultimately drives outbreak severity through transmission rate.

3-hop Causal Chains

For deeper analysis, find 3-hop chains ($A \rightarrow B \rightarrow C \rightarrow D$):

```
paths3 = causal_paths(csql; depth=3, limit=5)
println("Found $(length(paths3)) 3-hop paths:")
for p in paths3
    println(" $(p.a) -[$(p.r1)]→ $(p.b) -[$(p.r2)]→ $(p.c) -[$(p.r3)]→ $(p.d)")
    println(" path_score=$(round(p.path_score; digits=2))")
end
```

Found 5 3-hop paths:

```
pathogen transmissibility -[INCREASES]→ transmission rate -[INCREASES]→ outbreak severity
path_score=2.75
contact rate -[INCREASES]→ transmission rate -[INCREASES]→ outbreak severity -[INCREASES]→ public health response
path_score=2.7
transmission rate -[INCREASES]→ outbreak severity -[INCREASES]→ public health response -[INCREASES]→ case fatality rate
path_score=2.65
host susceptibility -[INCREASES]→ transmission rate -[INCREASES]→ outbreak severity -[INCREASES]→ public health response
path_score=2.63
quarantine -[REDUCES]→ transmission rate -[INCREASES]→ outbreak severity -[INCREASES]→ public health response
path_score=2.61
```

Feedback Loops

Detect 2-cycles — pairs of concepts that mutually influence each other:

```
loops = feedback_loops(csql)
println("Found $(length(loops)) feedback loops:")
for l in loops
    println("  $(l.a)  $(l.b)")
    println("    $(l.a) -[$(l.r1)]→ $(l.b)")
    println("    $(l.b) -[$(l.r2)]→ $(l.a)")
    println("    loop_score=$(round(l.loop_score; digits=2))")
end
```

Found 1 feedback loops:

```
public health response  outbreak severity
public health response -[REDUCES]→ outbreak severity
outbreak severity -[INCREASES]→ public health response
loop_score=1.7
```

Feedback loops are epidemiologically significant — they represent reinforcing or balancing dynamics in disease systems.

Controversial Claims

Find edges where evidence is mixed (some sources say “increases”, others say “reduces”):

```
controv = controversial_claims(csql; threshold=0.1)
println("Found $(length(controv)) controversial claims:")
for c in controv
    println("  $(c.src) -[$(c.rel_type)]→ $(c.dst)")
    println("    controversy=$(round(c.controversy; digits=3)), inc=$(round(c.pol_mass_inc; digits=3))")
end
```

Found 0 controversial claims:

A controversy score near 0.5 means evidence is evenly split between increase and decrease.

Relation Distribution

Understand what types of causal relationships dominate the atlas:

```
rel_dist = stats[:relation_distribution]
for r in rel_dist
    println("  $(r.rel_type): $(r.n) edges, total_mass=$(round(r.total_mass; digits=2))")
end

INCREASES: 8 edges, total_mass=6.51
REDUCES: 6 edges, total_mass=4.75
```

Combining Queries with Custom SQL

Find concepts that are both causes and effects (intermediate nodes in causal chains):

```
intermediates = custom_query(csql, """
    SELECT n.label_canon,
           n.deg_in AS incoming,
           n.deg_out AS outgoing,
           n.deg_in + n.deg_out AS total_degree
    FROM atlas_nodes n
    WHERE n.deg_in > 0 AND n.deg_out > 0
    ORDER BY total_degree DESC
""")
println("Intermediate concepts (both cause and effect):")
for row in intermediates
    println("  $(row.label_canon): in=$(row.incoming), out=$(row.outgoing)")
end
```

Intermediate concepts (both cause and effect):

```
transmission rate: in=5, out=1
outbreak severity: in=2, out=3
host susceptibility: in=2, out=1
public health response: in=1, out=1
population immunity: in=1, out=1
healthcare capacity: in=1, out=1
```

Summary

This vignette demonstrated:

- **Multi-hop paths** (`causal_paths`) for discovering indirect causal chains
- **Feedback loops** (`feedback_loops`) for detecting reinforcing/balancing dynamics
- **Controversial claims** (`controversial_claims`) for finding contested evidence
- **Custom SQL** for flexible ad-hoc analysis

Next: Counterfactual Reasoning covers do-cut interventions and causal inference.